

Physical Implementation Analysis

Design : Parameterizable Output-Stationary INT8 Systolic Array IP Block
for Tiled Matrix-Matrix Multiplication
with Ping-Pong Buffered SRAM
supporting Single-Layer Inference and Off-Chip-Assisted Training

PDK : GF180MCU (gf180mcu_fd_sc_mcu7t5v0, 7-track, 5V)

Target : Slot A (~1,248,806 μm^2)

Target Frequency : 25 MHz

Section 1 — Verified PDK Cell Areas

All values sourced directly from gf180mcu_fd_sc_mcu7t5v0 Liberty file.

Standard Cells Used

Cell	Function	Area (μm^2)
addf_1	Full Adder	72.4416
addh_1	Half Adder	39.5136
dffq_1	D Flip-Flop (no reset)	63.6608
dffrnq_1	D Flip-Flop (with reset)	74.6368
nand2_1	2-input NAND	10.9760
mux2_1	2-to-1 MUX	28.5376
mux4_1	4-to-1 MUX	70.2464
and2_1	2-input AND	17.5616

SRAM Macros Used

Macro	Capacity	Width (μm)	Height (μm)	Area (μm^2)
sram128x8	128 bytes	431.86	268.88	116,118.52

SRAM Timing (from Liberty file, tt corner, 25°C, 5V):

Parameter	Value	Notes
min_period	5.88 ns	Supports up to ~170 MHz. Our 25 MHz target (40 ns period) is well within spec.
CLK→Q delay	~4.5 ns	Q valid ~4.5 ns after rising edge. Stable 35.5 ns before next edge at 25 MHz.
Read latency	2 cycles	Pipelined: addr at cycle N, Q valid at N+1, staging captures at N+2.
Write latency	1 cycle	Data, addr, write_en stable before rising edge → captured at that edge.

Section 2 — MAC Unit

One MAC unit computes:

`accumulator += A[7:0] × B[7:0]`
where A and B are signed INT8 elements of the two input matrices being multiplied.

2.1 — 8×8 Signed Multiplier (Baugh-Wooley, NAND-optimized)

Baugh-Wooley algorithm handles two's complement signed multiplication natively. NAND2 gates replace AND2 for partial product generation since nand2_1 (10.9760 μm^2) is cheaper than and2_1 (17.5616 μm^2).

Partial Product Generation:

$$\begin{aligned} 64 \times \text{nand2_1}: \\ 64 \times 10.9760 &= 702.46 \mu\text{m}^2 \end{aligned}$$

Wallace Tree Reduction:

$$\begin{aligned} \text{Stage 1: } 20 \times \text{addf_1} + 7 \times \text{addh_1} \\ &= (20 \times 72.4416) + (7 \times 39.5136) \\ &= 1,448.83 + 276.60 \\ &= 1,725.43 \mu\text{m}^2 \end{aligned}$$

$$\begin{aligned} \text{Stage 2: } 13 \times \text{addf_1} + 4 \times \text{addh_1} \\ &= (13 \times 72.4416) + (4 \times 39.5136) \\ &= 941.74 + 158.05 \\ &= 1,099.79 \mu\text{m}^2 \end{aligned}$$

$$\begin{aligned} \text{Stage 3: } 7 \times \text{addf_1} + 2 \times \text{addh_1} \\ &= (7 \times 72.4416) + (2 \times 39.5136) \\ &= 507.09 + 79.03 \\ &= 586.12 \mu\text{m}^2 \end{aligned}$$

$$\begin{aligned} \text{Final Carry-Propagate Adder: } 8 \times \text{addf_1} \\ &= 8 \times 72.4416 \\ &= 579.53 \mu\text{m}^2 \end{aligned}$$

Multiplier Total:

$$702.46 + 1,725.43 + 1,099.79 + 586.12 + 579.53 = 4,693.33 \mu\text{m}^2$$

2.2 — 21-bit Accumulator Adder

Accumulator width derivation:

$$\begin{aligned} \text{Maximum single INT8} \times \text{INT8 product:} \\ (-128) \times (-128) &= 16,384 = 2^{14} \rightarrow \text{fits in 15 bits unsigned} \end{aligned}$$

For 8×8 array with tiling up to 32×32 matrix:
Maximum accumulations per output element = 32
Maximum accumulated value = $32 \times 16,384 = 524,288 = 2^{19}$
Bits needed to represent $2^{19} = 20$ bits unsigned (bit 19 must be set)
With sign bit = 21 bits signed
21-bit signed range: $\pm 1,048,575 \rightarrow$ safely covers 524,288 

$$\begin{aligned} 21 \times \text{addf_1}: \\ 21 \times 72.4416 &= 1,521.27 \mu\text{m}^2 \end{aligned}$$

2.3 — 21-bit Accumulator Register

Using dffrnq_1 (DFF with built-in reset) instead of dffq_1 + mux2_1:

$$\begin{aligned} \text{Comparison:} \\ \text{dffq_1} + \text{mux2_1}: & 63.6608 + 28.5376 = 92.1984 \mu\text{m}^2 \text{ per bit} \\ \text{dffrnq_1}: & 74.6368 \mu\text{m}^2 \text{ per bit} \\ \text{Saving per bit:} & 17.5616 \mu\text{m}^2 \\ \text{Saving per 21-bit accumulator:} & 368.79 \mu\text{m}^2 \end{aligned}$$

$$21 \times \text{dffrnq_1}:$$

$$21 \times 74.6368 = 1,567.37 \mu\text{m}^2$$

Reset signal is shared across all MAC units (one wire) — no extra area.

2.4 — Input Flow Registers

In a systolic array, A values flow left→right and B values flow top→bottom through registered connections between cells.

8-bit A Register (activation flows left→right):

$$8 \times \text{dffq_1}:$$

$$8 \times 63.6608 = 509.29 \mu\text{m}^2$$

8-bit B Register (weight flows top→bottom):

$$8 \times \text{dffq_1}:$$

$$8 \times 63.6608 = 509.29 \mu\text{m}^2$$

2.5 — Enable Gating Logic

$$4 \times \text{nand2_1}:$$

$$4 \times 10.9760 = 43.90 \mu\text{m}^2$$

2.6 — MAC Unit Total Cell Area

Component	Cell Area (μm^2)
8×8 Baugh-Wooley Multiplier	4,693.33
21-bit Accumulator Adder	1,521.27
21-bit Accumulator Register (dffrnq_1)	1,567.37
8-bit A Flow Register	509.29
8-bit B Flow Register	509.29
Enable Gating	43.90
Total per MAC unit:	8,844.45 μm^2

Section 3 — Full 8×8 Systolic Array

$$64 \text{ MAC units:}$$

$$64 \times 8,844.45 = 566,044.80 \mu\text{m}^2$$

Section 4 — Output Processor

One output_processor is shared across all 64 MAC outputs. It serializes results one byte per cycle through the 8-bit output bus, with internal row and column counters to sequence through all ARRAY_SIZE² outputs.

4.1 — Output Selection MUX (1-of-64, 21 bits wide)

Three-level mux4_1 tree to select one of 64 accumulator outputs:

Level 1: 16 groups of 4 → 16 × mux4_1 per bit
 Level 2: 4 groups of 4 → 4 × mux4_1 per bit
 Level 3: 1 final select → 1 × mux4_1 per bit
 Total per bit: 21 × mux4_1

For 21-bit accumulator:

$$21 \times 21 \times \text{mux4_1} = 441 \times 70.2464 = 30,978.66 \mu\text{m}^2$$

4.2 — Saturation Clip Logic (21-bit → 8-bit)

Overflow detection + clamp to [-128, +127]:

$$\begin{aligned} &8 \times \text{nand2_1} + 8 \times \text{mux2_1} \\ &= (8 \times 10.9760) + (8 \times 28.5376) \\ &= 87.81 + 228.30 \\ &= 316.11 \mu\text{m}^2 \end{aligned}$$

4.3 — 8-bit Output Register

$$\begin{aligned} &8 \times \text{dffq_1} \\ &8 \times 63.6608 = 509.29 \mu\text{m}^2 \end{aligned}$$

4.4 — Output Row and Column Counters

Row counter (3-bit for ARRAY_SIZE=8 rows):

$$3 \times \text{dffq_1} = 3 \times 63.6608 = 190.98 \mu\text{m}^2$$

Column counter (3-bit for ARRAY_SIZE=8 columns):

$$3 \times \text{dffq_1} = 3 \times 63.6608 = 190.98 \mu\text{m}^2$$

These counters sequence through all 64 MAC outputs during the OUTPUT phase and also serve as MUX select signals for output selection.

4.5 — output_done Register

Single-cycle pulse register asserted after all ARRAY_SIZE² results transferred:

$$\begin{aligned} &1 \times \text{dffq_1} \\ &1 \times 63.6608 = 63.66 \mu\text{m}^2 \end{aligned}$$

4.6 — output_processor Total

Component	Cell Area (μm ²)
Output Selection MUX	30,978.66
Saturation Clip Logic	316.11
8-bit Output Register	509.29
Row Counter (3-bit)	190.98
Column Counter (3-bit)	190.98
output_done Register (1-bit)	63.66
Total output_processor:	32,249.68 μm ²

Section 5 — Feeder

The feeder module reads tile data from the active SRAM using pipelined reads (2-cycle latency), fills A and B staging registers, applies skew buffer delays to produce the diagonal wavefront pattern for the systolic array, generates single-cycle valid pulses every 16 cycles, self-stops after 128 reads, and manages the 14-cycle drain phase on the last tile. All sequencing is fully internal.

5.1 — A and B Staging Registers

Staging registers capture SRAM data (2 cycles after address presented due to pipelined read latency) and hold values until skew buffer shifts every 16 cycles.

A_stage: 8 registers × 8 bits each:
 $64 \times \text{dffq_1} = 64 \times 63.6608 = 4,074.29 \mu\text{m}^2$

B_stage: 8 registers × 8 bits each:
 $64 \times \text{dffq_1} = 64 \times 63.6608 = 4,074.29 \mu\text{m}^2$

5.2 — Skew Buffer

Row i has i chained 8-bit register stages. Each stage holds its value for 16 clock cycles before passing to the next stage (shifts every 16 cycles, not every clock cycle). This implements the diagonal skew required for output-stationary systolic computation.

Total stages per A: $0+1+2+3+4+5+6+7 = 28$ stages
Total stages per B: $0+1+2+3+4+5+6+7 = 28$ stages

A skew: 28 stages × 8 bits = $224 \times \text{dffq_1}$
 $224 \times 63.6608 = 14,260.02 \mu\text{m}^2$

B skew: 28 stages × 8 bits = $224 \times \text{dffq_1}$
 $224 \times 63.6608 = 14,260.02 \mu\text{m}^2$

5.3 — Internal Counters

Read counter (7-bit, counts 0→127, self-stops at 128):

Tracks SRAM read progress. Stops asserting read_en after address 127 presented. Resets on swap_pulse for next tile.

$7 \times \text{dffq_1}$:
 $7 \times 63.6608 = 445.63 \mu\text{m}^2$

Phase tracker (1-bit, A column vs B row phase):

Tracks whether feeder is currently reading A columns (cycles 1-8 of each 16-cycle block) or B rows (cycles 9-16).

$1 \times \text{dffq_1}$:
 $1 \times 63.6608 = 63.66 \mu\text{m}^2$

Inner step counter (3-bit, counts 0→ARRAY_SIZE-1):

Tracks current inner step k (0→7). Determines which column of A and row of B to read.

$3 \times \text{dffq_1}$:
 $3 \times 63.6608 = 190.98 \mu\text{m}^2$

Drain counter (4-bit, counts 14 drain cycles on last_pass):

After 128 reads on last tile (last_pass=1), feeder enters 14-cycle drain phase. This counter tracks progress.

$4 \times \text{dffq_1}$:
 $4 \times 63.6608 = 254.64 \mu\text{m}^2$

5.4 — Control Logic

Combinational logic for: read_en generation, valid pulse generation (single cycle every 16 cycles), drain phase detection and zero injection, drain_done pulse generation, swap_pulse restart handling:

$$\begin{aligned} &\sim 20 \times \text{nand2_1}: \\ &20 \times 10.9760 = 219.52 \text{ } \mu\text{m}^2 \end{aligned}$$

5.5 — Feeder Total

Component	FFs	Cell Area (μm^2)
A Staging Registers (8×8-bit)	64	4,074.29
B Staging Registers (8×8-bit)	64	4,074.29
A Skew Buffer (28 stages × 8-bit)	224	14,260.02
B Skew Buffer (28 stages × 8-bit)	224	14,260.02
Read Counter (7-bit, 0-127)	7	445.63
Phase Tracker (1-bit, A vs B)	1	63.66
Inner Step Counter (3-bit, 0-7)	3	190.98
Drain Counter (4-bit, 0-13)	4	254.64
Control Logic ($\sim 20 \times \text{nand2_1}$)	—	219.52
Total Feeder:	591	37,843.05 μm^2

Section 6 — Controller FSM and Counters

The controller is a pure FSM. It manages all module operations: generates write_addr and write_en for SRAM loading, asserts swap on tile_done (sole swap condition), simultaneously asserts ready_in permission pulse with swap, forwards last_pass to feeder, monitors drain_done and output_done, asserts clear after output phase. It does not touch the data path.

Note on write_en: write_en = valid_in (in LOAD states), NOT valid_in AND ready_in. ready_in is a tile-level permission signal only, not a per-byte handshake. This means all 128 bytes of a tile are written when the host drives valid_in=1, regardless of ready_in state (which goes LOW after the first byte).

6.1 — FSM State Register

$$\begin{aligned} &4 \text{ states} \rightarrow 2\text{-bit encoding}: \\ &2 \times \text{dffq_1}: \\ &2 \times 63.6608 = 127.32 \text{ } \mu\text{m}^2 \end{aligned}$$

FSM States:
IDLE
INITIAL_LOAD
COMPUTE_AND_LOAD
OUTPUT

6.2 — write_addr Counter (7-bit)

Generates write address for incoming tile bytes. Increments on every cycle where valid_in=1 in LOAD states (write_en=1). Resets to 0 on swap. Stalls when valid_in=0.

$$\begin{aligned} &7 \times \text{dffq_1}: \\ &7 \times 63.6608 = 445.63 \text{ } \mu\text{m}^2 \end{aligned}$$

6.3 — SRAM Swap Register (1-bit)

Tracks which SRAM is active (being read by feeder) and which is inactive (being written by host). Flips on tile_done (sole swap condition). Swap pulse goes simultaneously to memory and feeder.

$$\begin{aligned} 1 \times \text{dffq}_1: \\ 1 \times 63.6608 &= 63.66 \mu\text{m}^2 \end{aligned}$$

6.4 — ready_in Register (1-bit)

Holds the tile-level permission pulse state. Asserted HIGH same cycle as swap on non-last tiles (and after reset for first tile). Deasserted LOW when first byte of new tile arrives (valid_in=1 while ready_in=1). Stays LOW during OUTPUT state and after last tile.

$$\begin{aligned} 1 \times \text{dffq}_1: \\ 1 \times 63.6608 &= 63.66 \mu\text{m}^2 \end{aligned}$$

6.5 — output_en Register (1-bit)

Enables output_processor during OUTPUT state.

$$\begin{aligned} 1 \times \text{dffq}_1: \\ 1 \times 63.6608 &= 63.66 \mu\text{m}^2 \end{aligned}$$

6.6 — last_pass_f Register (1-bit)

Registered forward of last_pass input to feeder. Feeder uses this to trigger drain phase after completing 128 reads.

$$\begin{aligned} 1 \times \text{dffq}_1: \\ 1 \times 63.6608 &= 63.66 \mu\text{m}^2 \end{aligned}$$

6.7 — Next State and Output Logic

Combinational logic computing next FSM state and all control outputs (swap, clear, output_en, ready_in, write_en, feeder signals):

$$\begin{aligned} \sim 40 \times \text{nand2}_1: \\ 40 \times 10.9760 &= 439.04 \mu\text{m}^2 \end{aligned}$$

6.8 — Controller Total

Component	Cell Area (μm^2)
FSM State Register (2-bit, 4 states)	127.32
write_addr Counter (7-bit)	445.63
SRAM Swap Register (1-bit)	63.66
ready_in Register (1-bit)	63.66
output_en Register (1-bit)	63.66
last_pass_f Register (1-bit)	63.66
Next State + Output Logic ($\sim 40 \times \text{nand2}_1$)	439.04
Total Controller:	1,266.63 μm^2

Section 7 — Host Interface

The host_interface module handles all physical pin-level communication for Chipathon testing. It contains output enable (OE) control for the bidirectional pad_data bus and synchronization registers for incoming pad signals. This module is specific to the Chipathon tapeout and is not part of the reusable accelerator_core IP.

7.1 — Bidirectional Bus OE Control

pad_data is an 8-bit bidirectional bus. When valid_out=1 (chip sending results), the chip drives pad_data from data_out. Otherwise, the host drives pad_data. One mux2_1 per bit selects between driving and high-Z:

$$8 \times \text{mux2_1}:$$

$$8 \times 28.5376 = 228.30 \text{ } \mu\text{m}^2$$

$$\text{OE control logic}:$$

$$2 \times \text{nand2_1} = 2 \times 10.9760 = 21.95 \text{ } \mu\text{m}^2$$

7.2 — Input Synchronization Registers

Registers for incoming pad signals: pad_valid_in, pad_tile_done, pad_last_pass, pad_ready_out:

$$4 \times \text{dffq_1}:$$

$$4 \times 63.6608 = 254.64 \text{ } \mu\text{m}^2$$

7.3 — Output Registers

Registers for pad_ready_in and pad_valid_out:

$$2 \times \text{dffq_1}:$$

$$2 \times 63.6608 = 127.32 \text{ } \mu\text{m}^2$$

7.4 — host_interface Total

Component	Cell Area (μm^2)
Bidir Bus OE MUX (8 × mux2_1)	228.30
OE Control Logic (2 × nand2_1)	21.95
Input Sync Registers (4 × dffq_1)	254.64
Output Registers (2 × dffq_1)	127.32
Total host_interface:	632.22 μm^2

Section 8 — Total Standard Cell Area

Component	Cell Area (μm^2)
64 MAC Units	566,044.80
output_processor	32,249.68
feeder	37,843.05
controller	1,266.63
host_interface	632.22
Total standard cell area:	638,036.38 μm^2

Section 9 — Placed Area at Different Utilizations

Standard cells only (SRAMs are hard macros, fixed size):

At 65% utilization (conservative):
 $638,036.38 \div 0.65 = 981,594.43 \text{ } \mu\text{m}^2$

At 70% utilization (realistic for structured systolic array):
 $638,036.38 \div 0.70 = 911,480.54 \text{ } \mu\text{m}^2$

At 75% utilization (optimistic):
 $638,036.38 \div 0.75 = 850,715.17 \text{ } \mu\text{m}^2$

Section 10 — Hard Macro Area (Fixed, No Utilization Factor)

Two sram128x8 macros for ping-pong buffering:

SRAM_0 (sram128x8): holds current tile (64 bytes A + 64 bytes B) – read by feeder
SRAM_1 (sram128x8): holds next tile (written by host during compute phase)

Each sram128x8:
Width: 431.86 μm
Height: 268.88 μm
Area: 116,118.52 μm^2

Two macros:
 $2 \times 116,118.52 = 232,237.04 \text{ } \mu\text{m}^2$

Section 11 — Grand Total Design Area

	65% util	70% util	75% util
Standard cells:	981,594	911,481	850,715
Two sram128x8:	232,237	232,237	232,237
Grand total:	1,213,831	1,143,718	1,082,952
Slot A %:	97.2%	91.6%	86.7%

All three utilization targets fit within Slot A. 70% is recommended as the realistic target for this structured systolic array design.

11.1 — Accumulator Width Sensitivity Analysis

The accumulator width (ACCUM_WIDTH) is a key parameterizable variable that determines the maximum matrix inner dimension K supported. Since ACCUM_WIDTH affects both the MAC unit accumulator registers and the output selection MUX, each additional bit increases total design area.

Extra area per additional accumulator bit (at 70% utilization):

MAC units ($64 \times (\text{addf_1} + \text{dffrnq_1})$):
Raw cell area: $64 \times (72.4416 + 74.6368) = 9,413.02 \text{ } \mu\text{m}^2$
At 70% util: $9,413.02 \div 0.70 = 13,447.17 \text{ } \mu\text{m}^2$

Output MUX (21 × mux4_1 per bit):
 Raw cell area: 21 × 70.2464 = 1,475.17 μm²
 At 70% util: 1,475.17 ÷ 0.70 = 2,107.39 μm²

Total extra area per bit (at 70%):
 13,447.17 + 2,107.39 = 15,554.56 μm²

Baseline: 21-bit accumulator at 70% utilization = 1,143,718 μm²

Extra Bits	ACCUM_WIDTH	Max K	Application Coverage	Total Extra Area	New Total	Slot A %
0	21	32	ECG, basic biosignals	0 μm ²	1,143,718 μm ²	91.6% ✓
1	22	64	+ Gesture, anomaly, keyword spotting	15,555 μm ²	1,159,273 μm ²	92.8% ✓
2	23	128	+ Tiny image classification	31,109 μm ²	1,174,827 μm ²	94.1% ✓
3	24	256	+ Small speech recognition	46,664 μm ²	1,190,382 μm ²	95.3% ✓
4	25	512	+ Medium CNN	62,218 μm ²	1,205,936 μm ²	96.6% ✓
5	26	1,024	+ Large CNN, ResNet	77,773 μm ²	1,221,491 μm ²	97.8% ✓
6	27	2,048	+ BERT-small	93,327 μm ²	1,237,045 μm ²	99.1% ✓
7	28	4,096	+ GPT-2 small	108,882 μm ²	1,252,600 μm ²	100.3% ⚠
8	29	8,192	+ BERT-base	124,436 μm ²	1,268,154 μm ²	101.5% ⚠
9	30	16,384	+ GPT-2 medium	139,991 μm ²	1,283,709 μm ²	102.8% ⚠
10	31	32,768	+ GPT-3 small	155,546 μm ²	1,299,263 μm ²	104.0% ⚠

Key observations:

- Default ACCUM_WIDTH=21, targeting K≤32. Fits at 91.6% ✓
- All configurations up to ACCUM_WIDTH=27 (K≤2,048) fit within Slot A ✓
- ACCUM_WIDTH=28 and above exceeds Slot A at 70% utilization ⚠
- ACCUM_WIDTH is exposed as a top-level parameter allowing synthesis-time customization without any RTL changes ✓

Section 12 — Area Breakdown at 70% Utilization

Total Slot A: 1,248,806 μm²

Component	Cell Area (μm ²)	Placed Area (μm ²)	% of Slot A
64 MAC Units	566,045	808,636	64.8%
SRAM_0 (sram128x8)	116,119 (fixed)	116,119	9.3%
SRAM_1 (sram128x8)	116,119 (fixed)	116,119	9.3%
output_processor	32,250	46,071	3.7%
feeder	37,843	54,062	4.3%
controller	1,267	1,810	0.1%
host_interface	632	903	0.1%
Total	870,275	1,143,720	91.6%

Note: SRAM macros are hard macros — their placed area equals cell area (no utilization factor applied). Standard cells (MAC units, output_processor, feeder, controller, host_interface) are placed at 70% utilization.

Section 13 — Key Design Parameters Summary

Parameter	Value	Basis
Array size (parameterizable)	$\in \{2, 4, 8\}$	Constrained by sram128x8 capacity: $2 \times N^2 \leq 128$ bytes
Datapath precision	INT8 (signed)	Baugh-Wooley multiplier
Accumulator width (parameterizable)	≥ 21 bits (default 21)	Minimum for $K \leq 32$; increase for larger matrices
Accumulator register	dffrnq_1	Cheaper than dffq_1 + mux2_1
Multiplier type	Baugh-Wooley Wallace tree	NAND-optimized
SRAM configuration	$2 \times \text{sram128x8}$	Ping-pong buffering
SRAM tile capacity	128 bytes	64 bytes A + 64 bytes B per tile
SRAM read latency	2 cycles	Pipelined: addr@N, Q@N+1, staging@N+2
SRAM min period	5.88 ns	Supports up to ~170 MHz (tt, 25°C, 5V)
SRAM CLK→Q delay	~4.5 ns	Q stable 35.5 ns before next edge at 25 MHz
Target frequency	25 MHz	40 ns clock period, GF180MCU 5V
output_processor	1 shared	Serializes ARRAY_SIZE ² results, 1 per cycle
write_addr counter	In controller	7-bit, increments on valid_in=1 in LOAD states
Feeder FFs	591	Staging (128) + skew buffer (448) + counters (15)
Total std cell area	638,036 μm^2	PDK-verified cell areas
Total design area (70%)	1,143,718 μm^2	Realistic utilization
Slot A utilization (70%)	91.6%	Fits with 8.4% margin 

Section 14 — Important Caveats

- Slot A area taken as 1,248,806 μm^2 ($= 2235 \times 2235 \mu\text{m}$ total die area $\div 4$) as per Chipathon 2026 padframe proposal slides.
 - Cell counts are based on textbook decompositions. Yosys + ABC synthesis may achieve more efficient implementations using compound cells (AOI/OAI) from the library — potentially reducing area by 10–20%.
 - Utilization assumption of 70% is based on the regular, structured nature of systolic array designs. Actual utilization confirmed only after LibreLane place-and-route run.
 - FSM and control logic estimates (~40 gates) are approximations. Actual synthesis may vary.
 - SRAM timing parameters are sourced from the Liberty file at typical corner (tt), 25°C, 5V. Read latency is 2 clock cycles when using pipelined reads. At 25 MHz (40 ns period), Q settles ~4.5 ns after the rising edge and remains stable for ~35.5 ns before the next edge, providing ample setup margin.
 - The 8.4% free margin at 70% utilization provides sufficient buffer for routing overhead, filler cells, and any underestimation in cell counts.
 - Silicon validation required for final confirmation.
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Section 15 — Comparison: 4×4 vs 8×8

Metric	4×4 Array	8×8 Array
MAC units	16	64
Total std cell area	162,335 μm ²	638,036 μm ²
At 70% utilization	231,907 μm ²	911,481 μm ²
Two sram128x8	232,237 μm ²	232,237 μm ²
Grand total (70%)	464,144 μm ²	1,143,718 μm ²
Slot A utilization	37.2%	91.6%
Native matrix size	4×4	8×8
Compute cycles per tile	32	128
Load cycles per tile	32	128
Drain cycles (last tile)	6	14
Output cycles per tile	16	64
Throughput (MACs/cycle)	2	4
Throughput vs 4×4	1×	2×

Note: Both 4×4 and 8×8 figures now include feeder, controller, output_processor, and host_interface area for completeness. Drain cycles = 2×(ARRAY_SIZE-1).
